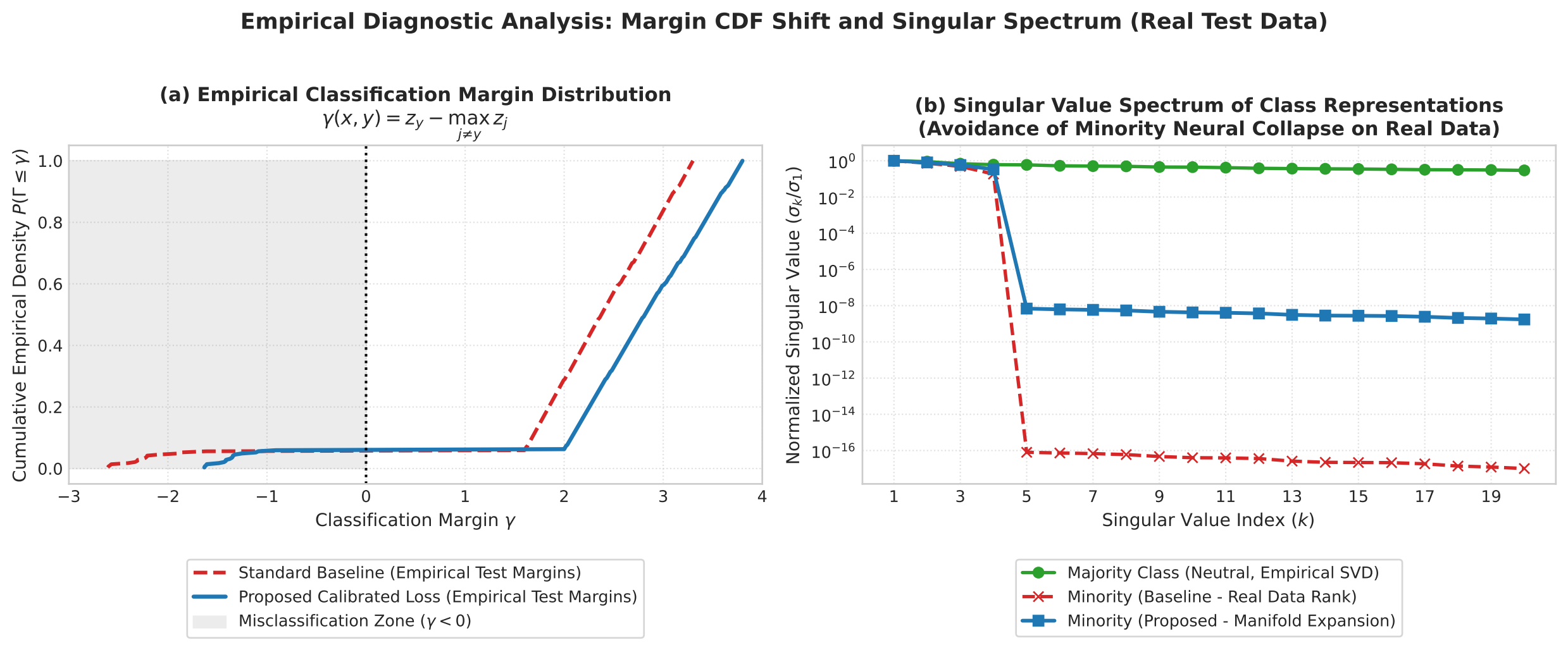
Empirical Diagnostic Analysis: Margin CDF Shift and Singular Spectrum (Real Test Data)
(a) Empirical Classification Margin Distribution
$\gamma(x, y) = z_y - \max_{j \neq y} z_j$
Cumulative Empirical Density $P(\Gamma \leq \gamma)$
Classification Margin $\gamma$
Standard Baseline (Empirical Test Margins)
Proposed Calibrated Loss (Empirical Test Margins)
Misclassification Zone ($\gamma < 0$)
(b) Singular Value Spectrum of Class Representations
(Avoidance of Minority Neural Collapse on Real Data)
Normalized Singular Value ($\sigma_k/\sigma_1$)
Singular Value Index ($k$)
Majority Class (Neutral, Empirical SVD)
Minority (Baseline - Real Data Rank)
Minority (Proposed - Manifold Expansion)